

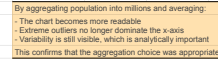
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Tokelau	NAN	1.35000024	0	99.70787665	0	0.252323459	0	99.70787665	0	0.252323459	0	NAN	NAN	NAN	NAN	0	NAN	0	99.70787665	99.70787665	0	0.252323459	0	
Niue	NAN	1.61800022	46.2020038	97.0108718	0	2.989123923	0	NAN	NAN	NAN	NAN	NAN	NAN	NAN	NAN	0.747584398	53.7979952	1	97.0108718	97.0108718	0	2.989123923	0	
Falkland Islands (Malvinas)	NAN	3.48300004	78.0579561	95.3089274	0	4.691027596	0	95.3089274	0	4.691027596	0	100	0	0	0	2.734543516	21.45200436	1	95.3089274	95.3089274	0	4.691027596	0	
Montserrat	NAN	4.99900072	91.0499771	98.0748262	0	1.922517378	0	NAN	NAN	NAN	NAN	NAN	NAN	NAN	NAN	0.4556858451	90.85500023	1	98.0748262	98.0748262	0	1.922517378	0	
Saint Pierre and Miquelon	NAN	5.795000076	89.96195799	91.4	0	8.6	0	NAN	NAN	NAN	NAN	NAN	NAN	NAN	NAN	5.213297852	10.03800201	1	91.4	91.4	0	8.6	0	
Saint Helena	NAN	6.071000099	40.0820073	99.1	0	0.9	0	NAN	NAN	NAN	NAN	NAN	NAN	NAN	NAN	2.433370034	59.91799527	1	99.1	99.1	0	0.9	0	
Saint Barthélemy	NAN	10.09199955	60.08799958	99.885	100	0	0	NAN	NAN	NAN	NAN	NAN	NAN	NAN	NAN	0	0	0	99.885	99.885	100	0	0	
Nauru	High income	10.83399963	100	100	0	0	0	NAN	NAN	NAN	NAN	100	0	0	0	10.83399963	0	1	100	100	0	0	0	
Wallis and Futuna Islands	NAN	11.24400029	0	99.14328736	0	0.856712639	0	99.14328736	0	0.856712639	0	NAN	NAN	NAN	NAN	0	100	1	99.14328736	99.14328736	0	0.856712639	0	
Tuvalu	Upper middle income	11.71999982	64.01399994	94.0	0	100	0	0	100	0	0	0	100	0	0	7.548530758	35.98600006	1	94.0	94.0	0	100	0	
Cook Islands	NAN	17.56399989	75.49500725	99.0710122	0	0.0263897827	0	NAN	NAN	NAN	NAN	NAN	NAN	NAN	NAN	13.25994168	24.50497525	1	99.0710122	99.0710122	0	0.0263897827	0	
Pitau	Upper middle income	18.09199955	60.86799958	99.65754055	0	0.3425446984	0	99.65754055	0	0.3425446984	0	NAN	NAN	NAN	NAN	14.852348	19.0200104	1	99.65754055	99.65754055	0	0.3425446984	0	
British Virgin Islands	High income	30.23699951	48.51469959	99.86436356	0	0.1326146384	0	99.86436356	0	0.1326146384	0	NAN	NAN	NAN	NAN	14.66903113	51.48500001	1	99.86436356	99.86436356	0	0.1326146384	0	
Gibraltar	High income	33.69100189	100	100	0	0	0	NAN	NAN	NAN	NAN	NAN	NAN	NAN	NAN	33.69100189	0	1	100	100	0	0	0	
San Marino	High income	33.93799973	97.49900055	100	0	0	0	NAN	NAN	NAN	NAN	NAN	NAN	NAN	NAN	33.08921054	2.50099945	1	100	100	0	0	0	
Liechtenstein	High income	38.13700104	14.41600337	100	0	0	0	NAN	NAN	NAN	NAN	NAN	NAN	NAN	NAN	5.497830211	85.58399963	1	100	100	0	0	0	
Saint Martin (French part)	High income	38.3869	99.9927139	100	0	0.0007286105507	0	NAN	NAN	NAN	NAN	NAN	NAN	NAN	NAN	99.9927139	0	1	100	100	0	0.0007286105507	0	
Monaco	High income	39.24399948	100	100	0	0	0	NAN	NAN	NAN	NAN	NAN	NAN	NAN	NAN	0	0	1	100	100	0	0	0	
Faeroe Islands	High income	48.86500168	42.39799881	100	0	0	0	NAN	NAN	NAN	NAN	NAN	NAN	NAN	NAN	20.71778283	57.60200119	1	100	100	0	0	0	
American Samoa	Upper middle income	55.19699986	87.22999988	99.73771166	0	0.2262283415	0	NAN	NAN	NAN	NAN	NAN	NAN	NAN	NAN	48.10584012	12.84700012	1	99.73771166	99.73771166	0	0.2262283415	0	
Greenland	High income	56.77199936	87.2800531	100.0000017	0	0	0	100	0	0	0	0	0	0	0	49.5517395	12.71799489	1	100	100	0	0	0	
Northern Mariana Islands	High income	57.50699921	91.79799562	100	0	0	0	NAN	NAN	NAN	NAN	NAN	NAN	NAN	NAN	58.6217213	8.22005448	1	100	100	0	0	0	
Marshall Islands	Upper middle income	59.19600024	77.73399199	93.76221631	11.387979	0.03997402585	0	94.42927	5.39028	0.18	0	NAN	NAN	NAN	NAN	49.0430237	22.36000891	1	93.76221631	93.76221631	11.387979	0.03997402585	0	
Bermuda	High income	62.27299981	100	99.90314002	0	0.0968598294	0	NAN	NAN	NAN	NAN	99.90314002	0	0.0968598294	0	62.27299981	0	1	99.90314002	99.90314002	0	0.0968598294	0	
Andorra	High income	77.26499939	87.91600037	100.0000037	0	0	0	100	0	0	0	0	100	0	0	67.92829715	12.08399963	1	100	100	0	0	0	
Isle of Man	High income	85.03199708	52.89800262	98.0705	0	0.925	0	NAN	NAN	NAN	NAN	NAN	NAN	NAN	NAN	44.98022636	47.01997338	1	98.0705	98.0705	0	0.925	0	
United States Virgin Islands	High income	104.4229995	90.9300209	98.18267384	0	1.281726284	0	98.18267384	0	1.281726284	0	NAN	NAN	NAN	NAN	100.4828217	4.06909701	1	98.18267384	98.18267384	0	1.281726284	0	
Tonga	Upper middle income	105.6869998	98.73100083	93.74800012	0.8814072962	0.3874218777	0	98.46807584	1.103159706	0.4287617551	0	99.60658149	0.143504536	0.2498012758	0	24.1448867	78.90100098	1	93.74800012	93.74800012	0.8814072962	0.3874218777	0	
Kiribati	Lower middle income	115.4459991	55.59309796	77.97902085	0.476896925	17.95217953	0	60.99418489	2.066099799	36.90250532	0	91.5117485	5.658666461	2.810158991	0	64.4604063	44.46000204	1	77.97902085	77.97902085	0.476896925	17.95217953	0	
Guam	High income	168.7830048	94.93800354	99.6952	0	0.3048	0	NAN	NAN	NAN	NAN	NAN	NAN	NAN	NAN	100.2392151	5.08199646	1	99.6952	99.6952	0	0.3048	0	
Saint Lucia	Upper middle income	183.628978	18.8409996	96.88784457	1.15652166	1.29503269	0	96.81402383	1.734152255	1.451823918	0	97.20582129	2.160718688	0.6274589237	0	35.94738744	81.1590004	1	96.88784457	96.88784457	1.15652166	1.29503269	0	
Samoa	Lower middle income	198.4100037	17.88899984	91.83772455	6.52099952	1.41862348	0.2227520679	91.78032877	6.22703339	1.72762816	62.1110006	0	35.4935584	62.1110006	0	35.4935584	62.1110006	1	91.83772455	91.83772455	6.52099952	1.41862348	0.2227520679	
San Tome and Principe	High income	209.1609995	28.795993	98.22645516	20.23794926	0.4435189284	1.09241919	74.33061449	19.71184258	1.72940474	4.258137229	79.80594229	20.1949371	0	0	162.5647952	25.45999809	1	98.22645516	98.22645516	20.23794926	0.4435189284	1.09241919	
Mayotte	NAN	272.8125883	45.75009465	96.37119771	0	3.628820295	NAN	NAN	NAN	NAN	NAN	NAN	NAN	NAN	NAN	124.8164688	54.24000055	1	96.37119771	96.37119771	0	3.628820295	0	
French Polynesia	High income	280.9039917	61.9750229	100	0	0	0	NAN	NAN	NAN	NAN	NAN	NAN	NAN	NAN	174.092553	38.02499771	1	100	100	0	0	0	
New Caledonia	High income	285.4909973	71.5179774	99.30532057	0	0.8846794278	0	NAN	NAN	NAN	NAN	NAN	NAN	NAN	NAN	204.177445	28.48200228	1	99.30532057	99.30532057	0	0.8846794278	0	
Barbados	High income	287.3710022	31.19099988	98.51445042	0.267800761	1.217949005	0	NAN	NAN	NAN	NAN	NAN	NAN	NAN	NAN	99.63389284	68.80900022	1	98.51445042	98.51445042	0.267800761	1.217949005	0	
French Guiana	High income	295.6820004	85.8199999	93.76221631	0	0.217783652	0	NAN	NAN	NAN	NAN	NAN	NAN	NAN	NAN	256.2382765	14.38000031	1	93.76221631	93.76221631	0	0.217783652	0	
Vanuatu	Lower middle income	307.1469939	25.52500153	91.23119075	1.06238505	0	7.706424022	88.39702133	1.255138275	0	0	10.3470661	99.5	0.5	0	78.40000494	74.47499847	1	91.23119075	91.23119075	1.06238505	0	7.706424022	
Iceland	High income	341.25	83.897995	99.9999721	0	0.000002794650015	0	99.9999721	0	0.000002794650015	0	0	0	0	0	320.4280006	6.1200005	1	99.9999721	99.9999721	0	0.000002794650015	0	
Maritine	NAN	375.2650146	89.13970788	94.81970788	0	0.1580292158	0	NAN	NAN	NAN	NAN	NAN	NAN	NAN	NAN	334.511207	10.86000001	1	94.81970788	94.81970788	0	0.1580292158	0	
Belize	Upper middle income	387.6371022	46.02500153	98.40195483	1.249110629	0.3489347411	0	97.99527058	1.358317682	0.6464747393	0	98.877968	1.12104	0	0	103.050723	53.7499847	1	98.40195483	98.40195483	1.249110629	0.3489347411	0	
Guadeloupe	High income	403.1270142	96.49899922	99.80312604	0	0.198973961	0	99.80312604	0	0.198973961	0	NAN	NAN	NAN	NAN	396.591274	1.50100708	1	99.80312604	99.80312604	0	0.198973961	0	
Brunei Darussalam	High income	437.4830017	78.25007763	99.9000368	0	0.09696320036	0	NAN	NAN	NAN	NAN	99.65	0	0.35	0	342.3304622	21.74999237	1	99.9000368	99.9000368	0	0.09696320036	0	
Malta	High income	441.5390015	94.7440033	100.000004	0	0	0	100	0	0	0	0	100	0	0	418.3317282	5.2559967	1	100	100	0	0	0	
Maldives	Upper middle income	504.5419922	40.68899872	99.5444386	0.048050913	0.407510705	0	99.9190125	0.0800875	0	0	98.9879883	0	0	0	219.531029	59.33100128	1	99.5444386	99.5444386	0.048050913	0.407510705	0	
Cabo Verde	Lower middle income	555.987991	68.65200483	98.76960462	7.904150846	3.209147716	0	11.07995414	80.11425544	0.911271022	9.63217003	0.351130397	93.1	6.9	0	50.3771082	33.34799987	1	98.76960462	98.76960462	7.904150846	3.209147716	0	
Suriname	High income	560.937792	66.14902828	99.9999991	0.563749847	0.0537848827	0	99.9999991	0.563749847	0.0537848827	0	99.1385387	0.7844652271	0.499118072	0	308.052156	33.85999792	1	99.9999991	99.9999991	0.563749847	0.0537848827	0	
Luxembourg	High income	625.9790132	61.95129563	99.87952354	0	0.1300074649	0	98.9525281	0	1.404761905	0	0	98.9525281	0	0	572.478138	8.5470047	1	99.87952354	99.87952354	0	0.1300074649	0	
Montenegro	Upper middle income	628.0620117	67.48800959	98.85691652	0.5454788204	0.99231200																		

Overall conclusion

When the urban share increases, the rural share decreases

This confirms the logical relationship:
Urban share + Rural share = 100%

This validates that the data and transformations were done correctly.



As population size increases:

- Urban and rural shares fluctuate significantly
 - There is no clear upward or downward trend
 - Countries with similar population sizes can have very different urbanization levels
- Population size alone does not determine how urbanized a country is.

Across small, medium, and large population groups:

- Some countries are highly urbanized
 - Others remain predominantly rural
 - This variability persists even among very large populations
- Urbanization depends more on economic structure, development

Overall conclusion

The visualization shows an inverse relationship between urban and rural population shares, as expected. However, no clear trend emerges between national population size and the degree of urbanization. Countries with similar population sizes can have very different urban-rural distributions, indicating that population size alone does not explain urbanization patterns.

Investigating access by area

	was bas r	was lin r	was usmp r	was usmp r	was bas r	was usmp r	was usmp r	was bas r	was lin r	was usmp r	was usmp r	was usmp r
MAX	37.422996287	33.5913717	30.36979308		100	42.16438068	51.21598167	40.51813242	30.24777809	19.83580881	36.32106505	
75TH PERCENTILE	27.20244205	29.4647478	20.91762634		100	61.96721683	40.51813242	40.48186495				
50TH PERCENTILE	19.4647478	21.85626534	4.41901047	5.19188504	81.33591685	5.89365254	8.72226144	2.42899373	3.38116935	1.71787604	0.31311484	
MEAN	37.3841397	30.47021861	0.63564740	0.17371663	81.12303635	3.22047913	4.66077035	98.10662849	0.05	0.3948740029		
MODE	100	100	100	100	100	100	100	100	100	100	100	100
MIN	85.8433106	0	0.03296022	0.62426111	86.42426111	0.16481040	0.56238171	0.0	0	0	0	0
Standard Deviation	99.85752154	4.85121957	0.237090022	1.876031993	99.11024429	8.553171216	13.2663303	1.57471481	99.94180183	3.966347184	2.059001302	0.159766056
1ST QUANTILE	14.24421428	4.85121957	0.254403699	1.876031993	94.23062139	8.553171216	13.1702638	1.57471481	97.38650451	3.966347184	2.059001302	0.159766056
Standard Error	0.05887616	0.96048653	0.714477036	3.923975937	21.50665741	8.70476658	11.82849434	8.84795138	0.15887616	6.55305705	3.25457302	0.869631624

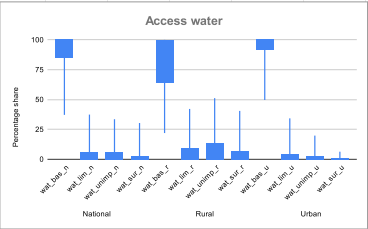
We observe that:
Mean $\neq$ Median $\neq$ Mode for all four features
this tells us that:
The distributions are not normal (not symmetric).
This is expected in global data because:
The countries are very heterogeneous
The access levels are clustered at extremes (near 0% or near 100%)

Interpretations:

National share
of population
with at least
basic drinking
water access

Most countries have very high national access to basic drinking water, and many have full coverage. However, a small group of countries with very low access creates inequality at the global level. This is why the mean is lower than the median and why variability exists despite high overall access.

National share of population with limited drinking water service	wat_lim_n	In most countries, limited water access is very low or zero, which is why the median is close to 0 and the most common value is 0. However, a small group of countries still has significant limited access, raising the mean and increasing the spread.
National share of population with unimproved drinking water service	wat_unimp_n	Most countries have unimproved water access that is very low or near zero, which is why the median is small and the mode is 0. However, a smaller group of countries still has high unimproved access, pushing the mean upward and creating noticeable variability across the dataset.
National share of population using surface water for drinking water	wat_sur_n	Surface water use is zero for most countries, which is why the median and mode are 0 and the IQR is small. However, a few countries still rely heavily on surface water, which increases the maximum and makes the standard deviation larger than what the median alone suggests.
Rural share of population with at least basic drinking water access	wat_bas_r	Most countries have high rural basic water access and many are near 100%. However, a minority of countries have very low rural access, which creates strong inequality and explains the large IQR and standard deviation.
Rural share of population with limited drinking water service	wat_lim_r	Most countries have very low (often zero) rural limited access, but a minority of countries still has high limited service in rural areas, which raises the mean and increases variability.
Rural share of population with unimproved drinking water service	wat_unimp_r	Most countries have low rural unimproved access, but a minority of countries still has very high unimproved water use in rural areas. This is why the mean is higher than the median and why IQR/STD are relatively large.
Rural share of population using surface water for drinking water	wat_sur_r	Most countries have very low (often near zero) rural surface water use, but a few countries rely heavily on surface water in rural areas, which raises the maximum and increases variability.
Urban share of population with at least basic drinking water access	wat_bas_u	Urban basic water access is very high for most countries and often reaches 100%. The distribution is tight (small IQR and STD), meaning urban access is relatively similar across countries, with a few low-access exceptions.
Urban share of population with limited drinking water service	wat_lim_u	Urban limited access is usually very low or zero, but a few countries still show high limited service in urban areas, raising the maximum and increasing spread.
Urban share of population with unimproved drinking water service	wat_unimp_u	Urban unimproved access is near zero for most countries, but a few countries still have noticeable unimproved access, which increases the maximum and spread.
Urban share of population using surface water for drinking water	wat_sur_u	Urban surface water use is essentially zero for most countries (median mode = 0, tiny IQR). Only a small number of countries have noticeable urban surface water reliance, which explains the maximum and the standard deviation.

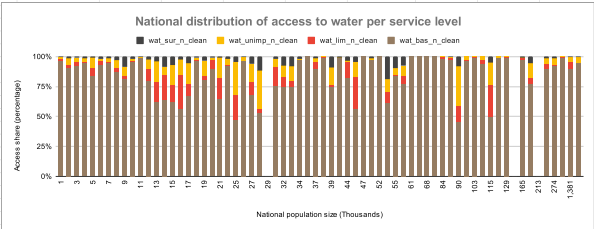


Interpretation by service type			
At least basic access	National (wat_bas_n)	box high (near the top) → most countries have high national basic access, but some lower outliers exist.	Urban > National > Rural in basic access, and rural has the largest inequality
	Rural (wat_bas_r)	box lower and much wider → rural basic access varies a lot; some countries are far behind.	
	Urban (wat_bas_u)	box highest and tightest → urban basic access is near universal in most countries.	
Limited access	Urban (wat_lim_u)	box close to 0 → in most countries, urban limited access is almost zero.	limited access is generally low, but it appears more in rural areas and is concentrated in certain countries.
	National (wat_lim_n)	also close to 0 but slightly higher spread than urban.	
	Rural (wat_lim_r)	higher box and longer whiskers → a subset of countries still has meaningful rural limited access.	
Unimproved access	Urban (wat_unimp_u)	box very close to 0 → unimproved access is rare in urban areas for most countries.	unimproved access is mostly a rural issue, with strong cross-country differences.
	National (wat_unimp_n)	low but slightly higher than urban.	
	Rural (wat_unimp_r)	clearly higher box and long upper whisker → rural unimproved access is a major problem in some countries.	
Surface water	Urban (wat_sur_u)	almost all values near 0 → urban surface water use is nearly nonexistent.	surface water use is generally rare, but when it exists, it is mostly rural and concentrated in a few countries.
	National (wat_sur_n) Rural (wat_sur_r)	Near 0 but with some countries higher. higher and more spread → some countries rely heavily on surface water in rural areas.	

Overall pattern across areas	
Urban is best and most consistent	For wat_bas_u, the box is very high and relatively tight → most countries have very high urban basic access with limited variation.
Rural shows the biggest inequality	For wat_bas_r, the box is lower and much taller than urban → rural basic access is lower and much more unequal across countries. For "problem" indicators (wat_lim_r, wat_unimp_r, wat_sur_r), rural boxes and whiskers are generally higher → rural areas have more limited, unimproved, and surface water use.
National sits between Urban and Rural	National distributions usually fall between urban and rural, which makes sense because national values combine both.

The box-and-whisker plots show clear differences by area. Urban basic water access is consistently high across countries, with a tight distribution close to 100%. Rural basic access is lower and much more variable, indicating stronger inequality between countries. For limited, unimproved, and surface water indicators, distributions are concentrated near zero in urban areas but are higher and more dispersed in rural areas, meaning these problems are primarily rural and concentrated in a subset of countries.

Investigating access by population size



- Most population-size bins will be dominated by basic access (large blue section).
- The remaining sections (limited/unimproved/surface) appear as smaller bands.
- We don't see a strong "population size → access" pattern; differences are more influenced by development/income and rural/urban structure.

name	pop_u	pop_u (Rounded)	wat_bas_u	wat_lim_u	wat_unimp_u	wat_sur_u	wat_bas_u_clean	wat_lim_u_clean	wat_unimp_u_clean	wat_sur_u_clean			
Tokelau	0		0	NAN	NAN	NAN	NAN						
Wallis and Futuna Islands	0		0	NAN	NAN	NAN	NAN						
Montserrat	9.114999771		9	NAN	NAN	NAN	NAN						
Papua New Guinea	13.34500027		13	86.07527164	0.1721822608	7.390385564	6.362160532	86.07527164	0.1721822608	7.390385564	6.362160532		
Liechtenstein	14.41600037		14	NAN	NAN	NAN	NAN						
Burundi	13.70800018		14	90.66911086	8.034462869	1.296426269	0	90.66911086	8.034462869	1.296426269	0		
Rwanda	17.43200111		17	83.0175993	9.250506251	5.462303057	2.269591394	83.0175993	9.250506251	5.462303057	2.269591394		
Malawi	17.42700005		17	86.45469821	10.27894825	2.897503996	0.3688495446	86.45469821	10.27894825	2.897503996	0.3688495446		
Niger	16.62599945		17	85.81383976	9.950440829	2.13468778	2.101031634	85.81383976	9.950440829	2.13468778	2.101031634		
Samoa	17.88899994		18	92.10119	7.89881	0	0	92.10119	7.89881	0	0		
Saint Lucia	18.84099996		19	97.20582219	2.166718886	0.6274589237	0	97.20582219	2.166718886	0.6274589237	0		
Sri Lanka	18.71299934		19	99.56938019	0.1273977901	0.3032220202	0	99.56938019	0.1273977901	0.3032220202	0		
South Sudan	20.19899994		20	70.01770445	18.71066092	11.27163464	0	70.01770445	18.71066092	11.27163464	0		
Nepal	20.57600021		21	89.57763316	3.101581411	6.359123168	0.9616622585	89.57763316	3.101581411	6.359123168	0.9616622585		
Ethiopia	21.69499969		22	84.21282816	14.30300858	1.091582355	0.392580896	84.21282816	14.30300858	1.091582355	0.392580896		
Tonga	23.09899902		23	99.60665819	0.143540536	0.2498012758	0	99.60665819	0.143540536	0.2498012758	0		
Eswatini	24.17100143		24	96.74976305	0.7460644731	1.632164244	0.8720082294	96.74976305	0.7460644731	1.632164244	0.8720082294		
Chad	23.52000046		24	74.19222309	15.99651381	8.922452721	0.8888103843	74.19222309	15.99651381	8.922452721	0.8888103843		
Cambodia	24.23200035		24	90.45807717	8.85091472	0	0.6910081144	90.45807717	8.85091472	0	0.6910081144		
Solomon Islands	24.67000008		25	91.41077689	3.557120425	3.899286137	1.132816553	91.41077689	3.557120425	3.899286137	1.132816553		
Uganda	24.95400047		25	78.7828518	13.74416783	6.421044133	1.051936238	78.7828518	13.74416783	6.421044133	1.051936238		
Vanuatu	25.52500153		26	99.5	0.5	0	0	99.5	0.5	0	0		
Afghanistan	26.02599907		26	100	0	0	0	100	0	0	0		
Guyana	26.7859993		27	100	0	0	0	100	0	0	0		
Tajikistan	27.50599861		28	95.58062386	0.8848144763	1.527268323	2.007293339	95.58062386	0.8848144763	1.527268323	2.007293339		
Kenya	27.99499893		28	86.97523145	4.327944725	2.73514263	5.961681193	86.97523145	4.327944725	2.73514263	5.961681193		
Lesotho	29.02799988		29	93.02090855	2.698211997	3.625530038	0.65534942	93.02090855	2.698211997	3.625530038	0.65534942		
Barbados	31.19099998		31	NAN	NAN	NAN	NAN						
Timor-Leste	31.31999969		31	96.47695792	1.527338152	1.995703929	0	96.47695792	1.527338152	1.995703929	0		
Burkina Faso	30.60700035		31	80.08191854	14.60004919	4.875836487	0.4421957805	80.08191854	14.60004919	4.875836487	0.4421957805		
Myanmar	31.14100075		31	95.42788606	0	1.830283006	2.741830931	95.42788606	0	1.830283006	2.741830931		
Zimbabwe	32.24200058		32	92.89512895	4.964517757	2.122298674	0.01805461538	92.89512895	4.964517757	2.122298674	0.01805461538		
Sudan	35.25299835		35	73.76365956	25.22343986	0	1.012900581	73.76365956	25.22343986	0	1.012900581		
United Republic of Tanzania	35.22700119		35	88.7939249	6.337231166	2.194178786	2.674665152	88.7939249	6.337231166	2.194178786	2.674665152		
India	34.9260025		35	93.6700363	3.272056027	2.820607523	0.2373001538	93.6700363	3.272056027	2.820607523	0.2373001538		
Lao People's Democratic Republic	36.29000092		36	97.06426222	0	2.935737778	0	97.06426222	0	2.935737778	0		
Kyrgyzstan	36.85599899		37	99.30769231	0.6923076923	0	0	99.30769231	0.6923076923	0	0		
Guinea	36.875		37	86.58169007	12.95724153	0.461068398	0	86.58169007	12.95724153	0.461068398	0		
Mozambique	37.0739975		37	87.98504792	5.452291027	4.912684644	1.649976404	87.98504792	5.452291027	4.912684644	1.649976404		
Viet Nam	37.34000015		37	99.18304001	0	0.816959987	0	99.18304001	0	0.816959987	0		
Pakistan	37.16500092		37	92.76804985	3.771561162	3.126184571	0.3342044223	92.76804985	3.771561162	3.126184571	0.3342044223		
Yemen	37.90799713		38	77.01412934	21.48137937	1.504491286	0	77.01412934	21.48137937	1.504491286	0		
Bangladesh	38.17700195		38	97.40277797	1.639520121	0.6993752147	0.2583266934	97.40277797	1.639520121	0.6993752147	0.2583266934		
Madagascar	38.5340004		39	80.47000694	4.498430298	13.74049434	1.291068417	80.47000694	4.498430298	13.74049434	1.291068417		
Saint Helena	40.08200073		40	NAN	NAN	NAN	NAN						
Maldives	40.66899872		41	98.99798893	0	1.002011069	0	98.99798893	0	1.002011069	0		
Mauritius	40.75999832		41	99.92231522	0	0.07768478119	0	99.92231522	0	0.07768478119	0		
Faeroe Islands	42.39799881		42	NAN	NAN	NAN	NAN						
Bhutan	42.31599808		42	98.10662849	1.3591415	0.4169208475	0.1173091586	98.10662849	1.3591415	0.4169208475	0.1173091586		
Central African Republic	42.19799805		42	49.66166495	34.27978009	15.95940214	0.09915281712	49.66166495	34.27978009	15.95940214	0.09915281712		
Republic of Moldova	42.84900284		43	97.35744448	1.516060408	1.126495111	0	97.35744448	1.516060408	1.126495111	0		
Sierra Leone	42.92300034		43	78.41163766	14.07927743	5.485735437	2.023349475	78.41163766	14.07927743	5.485735437	2.023349475		
Togo	42.79999924		43	90.58991001	3.229348056	5.307990528	0.8727514049	90.58991001	3.229348056	5.307990528	0.8727514049		
Egypt	42.78300095		43	99.5837311	0.1061012067	0.3101676943	0	99.5837311	0.1061012067	0.3101676943	0		
Guinea-Bissau	44.19599915		44	70.61434791	20.03063358	9.340625284	0.01439322693	70.61434791	20.03063358	9.340625284	0.01439322693		
Mali	43.90900421		44	95.91745475	3.947121723	0.1354235314	0	95.91745475	3.947121723	0.1354235314	0		
Zambia	44.6289978		45	86.73385466	3.451996194	9.130414105	0.6837350463	86.73385466	3.451996194	9.130414105	0.6837350463		
Niue	46.20200348		46	NAN	NAN	NAN	NAN						
Mayotte	45.75099945		46	NAN	NAN	NAN	NAN						
Belize	46.02500153		46	98.87896	1.12104	0	0	98.87896	1.12104	0	0		
Somalia	46.14099884		46	79.12330574	17.26991556	3.606778703	0	79.12330574	17.26991556	3.606778703	0		
Democratic Republic of the Congo	45.63800049		46	74.50335478	14.34445818	10.29905862	0.8531284131	74.50335478	14.34445818	10.29905862	0.8531284131		

Only include rows where the four categories are present, so every bar is a valid full composition.

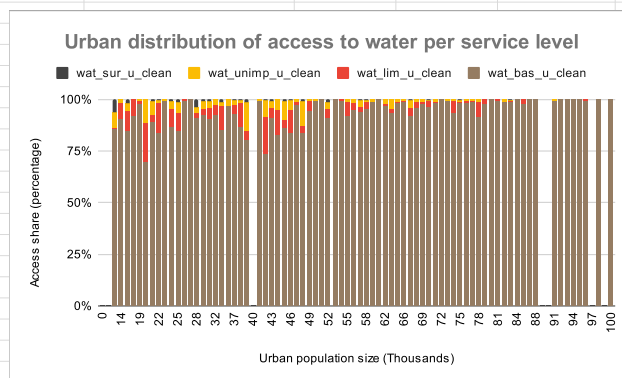
Philippines	47.40799713	47	97.4893325	1.632836465	0.877831036	0	97.4893325	1.632836465	0.877831036	0		
Benin	48.4149971	48	73.25800004	5.708056353	19.83580681	1.198136794	73.25800004	5.708056353	19.83580681	1.198136794		
Senegal	48.12200165	48	95.32170094	0.5920836567	4.086215402	0	95.32170094	0.5920836567	4.086215402	0		
British Virgin Islands	48.51499939	49	NAN	NAN	NAN							
Bosnia and Herzegovina	49.02000046	49	94.8457044	5.026922206	0.1273733928	0	94.8457044	5.026922206	0.1273733928	0		
Uzbekistan	50.41599655	50	99.55600556	0	0.4439944434	0	99.55600556	0	0.4439944434	0		
Thailand	51.43000031	51	100	0	0	0	100	0	0	0		
Namibia	52.03300095	52	96.26696535	2.584758965	0.6596102638	0.4886654211	96.26696535	2.584758965	0.6596102638	0.4886654211		
Liberia	52.08899689	52	85.52810963	10.69005698	3.58408039	0.1977530006	85.52810963	10.69005698	3.58408039	0.1977530006		
Guatemala	51.83599854	52	97.61816944	0.2768660057	1.94525887	0.1597056849	97.61816944	0.2768660057	1.94525887	0.1597056849		
Côte d'Ivoire	51.70599747	52	85.09355329	4.818535156	5.80236656	4.285544994	85.09355329	4.818535156	5.80236656	4.285544994		
Nigeria	51.95800018	52	92.35770519	2.99188065	3.356908786	1.29350537	92.35770519	2.99188065	3.356908786	1.29350537		
Isle of Man	52.89800262	53	NAN	NAN	NAN							
Trinidad and Tobago	53.2140007	53	NAN	NAN	NAN							
Turkmenistan	52.51600266	53	100	0	0	0	100	0	0	0		
Slovakia	53.75999832	54	99.60509406	0.3949059374	0	0	99.60509406	0.3949059374	0	0		
Romania	54.19400024	54	100	0	0	0	100	0	0	0		
Slovenia	55.11800385	55	NAN	NAN	NAN							
Mauritania	55.32699585	55	89.25869465	9.428808083	1.31249727	0	89.25869465	9.428808083	1.31249727	0		
Syrian Arab Republic	55.47500229	55	95.40516436	4.233570117	0.361265525	0	95.40516436	4.233570117	0.361265525	0		
Kiribati	55.59399796	56	91.53117485	5.658666461	2.810158691	0	91.53117485	5.658666461	2.810158691	0		
Jamaica	56.31100082	56	95.39801591	2.944456223	1.317064322	0.34046355	95.39801591	2.944456223	1.317064322	0.34046355		
Serbia	56.44599915	56	94.87156989	4.783037865	0.3453922445	0	94.87156989	4.783037865	0.3453922445	0		
Azerbaijan	56.39700317	56	100	0	0	0	100	0	0	0		
Fiji	57.24700546	57	98.19424406	0	1.495409344	0.3103465922	98.19424406	0	1.495409344	0.3103465922		
Haiti	57.08799744	57	84.62262129	7.227673879	8.149704835	0	84.62262129	7.227673879	8.149704835	0		
Ghana	57.34899902	57	96.12725952	2.608342455	1.185613803	0.07878421819	96.12725952	2.608342455	1.185613803	0.07878421819		
Indonesia	56.64099884	57	97.58051207	0.6085098345	1.679615139	0.1313629588	97.58051207	0.6085098345	1.679615139	0.1313629588		
North Macedonia	58.48199844	58	97.93740299	1.727136954	0.3354600563	0	97.93740299	1.727136954	0.3354600563	0		
Croatia	57.55299759	58	100	0	0	0	100	0	0	0		
Honduras	58.35899734	58	99.83728281	0.1537015262	0.009015663296	0	99.83728281	0.1537015262	0.009015663296	0		
Kazakhstan	57.67100143	58	98.00073133	1.999268673	0	0	98.00073133	1.999268673	0	0		
Cameroon	57.56000519	58	82.08358486	13.02379677	3.850701351	1.041917019	82.08358486	13.02379677	3.850701351	1.041917019		
Georgia	59.4529953	59	99.44615394	0	0.538220742	0.01562531586	99.44615394	0	0.538220742	0.01562531586		
Nicaragua	59.01200104	59	97.40695661	0.1075523143	2.244007656	0.2414834229	97.40695661	0.1075523143	2.244007656	0.2414834229		
Austria	58.7480011	59	100	0	0	0	100	0	0	0		
Poland	60.04300308	60	99.94435484	0	0.05564516129	0	99.94435484	0	0.05564516129	0		
French Polynesia	61.97500229	62	NAN	NAN	NAN							
Albania	62.11199951	62	95.66380912	1.62808683	2.708104054	0	95.66380912	1.62808683	2.708104054	0		
Paraguay	62.18299484	62	99.88674854	0.1132514643	0	0	99.88674854	0.1132514643	0	0		
Democratic People's Republic of Korea	62.38100052	62	96.91351044	0.8801428799	2.206346677	0	96.91351044	0.8801428799	2.206346677	0		
China	61.71308899	62	97.11488267	0.1831784927	2.37993884	0.322	97.11488267	0.1831784927	2.37993884	0.322		
Gambia	62.58199692	63	87.96690427	3.873689522	8.026457278	0.1329489253	87.96690427	3.873689522	8.026457278	0.1329489253		
Armenia	63.31299973	63	99.95448122	0	0.04551878364	0	99.95448122	0	0.04551878364	0		
Tuvalu	64.01399994	64	100	0	0	0	100	0	0	0		
Ireland	63.65299988	64	96.97988603	0	3.020113966	0	96.97988603	0	3.020113966	0		
Ecuador	64.16600037	64	100	0	0	0	100	0	0	0		
Morocco	63.53199768	64	97.89674601	1.375477117	0.7277768759	0	97.89674601	1.375477117	0.7277768759	0		
Suriname	66.14900208	66	98.71385387	0.7944652371	0.4916808925	0	98.71385387	0.7944652371	0.4916808925	0		
Portugal	66.30999756	66	100	0	0	0	100	0	0	0		
Cabo Verde	66.65200043	67	93.1	6.9	0	0	93.1	6.9	0	0		
Montenegro	67.48800659	67	99.19173959	0.8082604128	0	0	99.19173959	0.8082604128	0	0		
Cyprus	66.82099915	67	99.72515812	0	0.2748418825	0	99.72515812	0	0.2748418825	0		
Angola	66.82499695	67	71.74314862	9.558837489	17.72203473	0.9759791629	71.74314862	9.558837489	17.72203473	0.9759791629		
South Africa	67.35400391	67	99.00126484	0.734785403	0.2639497547	0	99.00126484	0.734785403	0.2639497547	0		
Latvia	68.31500244	68	98.87692469	0.9987568151	0.1243184908	0	98.87692469	0.9987568151	0.1243184908	0		
Lithuania	68.04599762	68	100	0	0	0	100	0	0	0		
Panama	68.41400146	68	98.08889267	1.911107333	0	0	98.08889267	1.911107333	0	0		
Congo	67.82900238	68	87.09329616	10.37996439	2.517789997	0.008949452055	87.09329616	10.37996439	2.517789997	0.008949452055		
Estonia	69.22900391	69	99.98545421	0	0.01454579307	0	99.98545421	0	0.01454579307	0		
Mongolia	68.65699768	69	96.64596558	1.718335079	1.635699341	0	96.64596558	1.718335079	1.635699341	0		
Bolivia (Plurinational State of)	70.1230011	70	99.12211832	0.09051328947	0.5613453808	0.2260230112	99.12211832	0.09051328947	0.5613453808	0.2260230112		
Tunisia	69.56800079	70	99.12119763	0.8788023684	0	0	99.12119763	0.8788023684	0	0		

Ukraine	69.60800171	70	91.27727011	8.14845779	0.1090456533	0.4652264446	91.27727011	8.14845779	0.1090456533	0.4652264446			
Botswana	70.8769989	71	97.62498058	0.4366200322	1.788579159	0.1498202281	97.62498058	0.4366200322	1.788579159	0.1498202281			
Iraq	70.89298774	71	99.80873103	0.19126897	0	0	99.80873103	0.19126897	0	0			
Italy	71.03899384	71	NAN	NAN	NAN	NAN							
New Caledonia	71.51799774	72	NAN	NAN	NAN	NAN							
Hungary	71.94200134	72	100	0	0	0	100	0	0	0			
El Salvador	73.44400024	73	99.57087001	0	0.4291299944	0	99.57087001	0	0.4291299944	0			
Sao Tome and Principe	74.35400391	74	79.5805429	20.4194571	0	0	79.5805429	20.4194571	0	0			
Switzerland	73.91500092	74	100	0	0	0	100	0	0	0			
Czech Republic	74.06100464	74	99.90265093	0	0.0973490679	0	99.90265093	0	0.0973490679	0			
Algeria	73.73300171	74	96.00473586	3.628288591	0.3669755478	0	96.00473586	3.628288591	0.3669755478	0			
Cook Islands	75.49500275	75	NAN	NAN	NAN	NAN							
Russian Federation	74.75400543	75	98.83259961	0.2973899687	0.6700104236	0.2	98.83259961	0.2973899687	0.6700104236	0.2			
Bulgaria	75.68599701	76	99.5286704	0	0.4713296007	0	99.5286704	0	0.4713296007	0			
Iran (Islamic Republic of)	75.87400055	76	98.64448672	1.188006671	0.1675066077	0	98.64448672	1.188006671	0.1675066077	0			
Turkey	76.10500336	76	97.32481783	1.815385203	0.7217969634	0.138	97.32481783	1.815385203	0.7217969634	0.138			
West Bank and Gaza Strip	76.71899414	77	97.62696443	1.246217358	1.126818208	NAN							
Cuba	77.19400024	77	97.77557664	1.1403859	1.059840374	0.02419708411	97.77557664	1.1403859	1.059840374	0.02419708411			
Malaysia	77.15999603	77	99.14309591	0.2983242605	0.4685798319	0.09	99.14309591	0.2983242605	0.4685798319	0.09			
Germany	77.45300293	77	100	0	0	0	100	0	0	0			
Marshall Islands	77.79399109	78	86.9	13.1	0	0	86.9	13.1	0	0			
Brunei Darussalam	78.25000763	78	99.65	0	0.35	0	99.65	0	0.35	0			
Djibouti	78.06199646	78	84.13501493	15.51511107	0.3498740029	0	84.13501493	15.51511107	0.3498740029	0			
Peru	78.2970047	78	96.55984408	0.6423932356	2.636408693	0.1613539953	96.55984408	0.6423932356	2.636408693	0.1613539953			
Falkland Islands (Malvinas)	78.50799561	79	100	0	0	0	100	0	0	0			
Belarus	79.48300171	79	96.01087362	3.985572644	0.003553731977	0	96.01087362	3.985572644	0.003553731977	0			
Greece	79.71500397	80	100	0	0	0	100	0	0	0			
Palau	80.98799896	81	99.63420199	0	0.3657980078	0	99.63420199	0	0.3657980078	0			
Costa Rica	80.77099609	81	99.84965286	0.1503471429	0	0	99.84965286	0.1503471429	0	0			
Libya	80.69100189	81	NAN	NAN	NAN	NAN							
Spain	80.80999756	81	99.90794521	0	0.09205479452	0	99.90794521	0	0.09205479452	0			
Colombia	81.42499542	81	99.9381928	0.0618072	0	0	99.9381928	0.0618072	0	0			
Republic of Korea	81.41400146	81	NAN	NAN	NAN	NAN							
France	80.97499847	81	100	0	0	0	100	0	0	0			
Mexico	80.73099518	81	100	0	0	0	100	0	0	0			
Canada	81.56200409	82	99.25827766	0	0.7417223371	0	99.25827766	0	0.7417223371	0			
Norway	82.97399139	83	100	0	0	0	100	0	0	0			
Dominican Republic	82.54000092	83	98.03689158	0.2727920359	1.302073396	0.3882429922	98.03689158	0.2727920359	1.302073396	0.3882429922			
United States of America	82.66400146	83	99.92814447	0	0.0718555296	0	99.92814447	0	0.0718555296	0			
Saudi Arabia	84.28700256	84	NAN	NAN	NAN	NAN							
United Kingdom	83.90299988	84	100	0	0	0	100	0	0	0			
French Guiana	85.81999969	86	NAN	NAN	NAN	NAN							
Oman	86.27599335	86	94.7	5.3	0	0	94.7	5.3	0	0			
Finland	85.51700592	86	100	0	0	0	100	0	0	0			
Australia	86.24099731	86	99.965	0	0.035	0	99.965	0	0.035	0			
American Samoa	87.15299988	87	NAN	NAN	NAN	NAN							
Greenland	87.28200531	87	100	0	0	0	100	0	0	0			
New Zealand	86.6989975	87	100	0	0	0	100	0	0	0			
United Arab Emirates	87.04799652	87	NAN	NAN	NAN	NAN							
Brazil	87.07299805	87	99.82200131	0	0.1779986949	0	99.82200131	0	0.1779986949	0			
Andorra	87.91600037	88	100	0	0	0	100	0	0	0			
Denmark	88.11600494	88	100	0	0	0	100	0	0	0			
Sweden	87.97699738	88	99.84155768	0	0.1584423157	0	99.84155768	0	0.1584423157	0			
Chile	87.72699738	88	100	0	0	0	100	0	0	0			
Venezuela (Bolivarian Republic of)	88.27899933	88	NAN	NAN	NAN	NAN							
Martinique	89.13999939	89	NAN	NAN	NAN	NAN							
Lebanon	88.92499542	89	NAN	NAN	NAN	NAN							
Saint Pierre and Miquelon	89.96199799	90	NAN	NAN	NAN	NAN							
Bahrain	89.50600433	90	NAN	NAN	NAN	NAN							
Gabon	90.09200287	90	89.80653917	7.42998076	2.763480072	NAN							
Luxembourg	91.4529953	91	100	0	0	0	100	0	0	0			
Jordan	91.41799927	91	99.09321259	0.1013669408	0.7437104707	0.06171	99.09321259	0.1013669408	0.7437104707	0.06171			
Northern Mariana Islands	91.79799652	92	NAN	NAN	NAN	NAN							

Netherlands	92.23600006	92	100	0	0	0	100	0	0	0
Argentina	92.11100006	92	99.79042065	0	0.2095793501	0	99.79042065	0	0.2095793501	0
Japan	91.78199768	92	NAN	NAN	NAN	NAN				
Israel	92.58699799	93	100	0	0	0	100	0	0	0
Iceland	93.897995	94	100	0	0	0	100	0	0	0
Puerto Rico	93.58100128	94	NAN	NAN	NAN	NAN				
Guam	94.93800354	95	NAN	NAN	NAN	NAN				
Malta	94.7440033	95	100	0	0	0	100	0	0	0
United States Virgin Islands	95.93900299	96	NAN	NAN	NAN	NAN				
Uruguay	95.51499939	96	99.69273444	0.3072655617	0	0	99.69273444	0.3072655617	0	0
San Marino	97.49900055	97	NAN	NAN	NAN	NAN				
Guadeloupe	98.49899292	98	NAN	NAN	NAN	NAN				
Belgium	98.07899475	98	100	0	0	0	100	0	0	0
Qatar	99.23500061	99	NAN	NAN	NAN	NAN				
Saint Barthelemy	100	100	100	0	0	0	100	0	0	0
Nauru	100	100	100	0	0	0	100	0	0	0
Gibraltar	100	100	100	0	0	0	100	0	0	0
Saint Martin (French part)	100	100	99.99927139	0	0.0007286105507	0	99.99927139	0	0.0007286105507	0
Monaco	100	100	100	0	0	0	100	0	0	0
Bermuda	100	100	99.90314002	0	0.09685998294	0	99.90314002	0	0.09685998294	0
China, Macao SAR	100	100	100	0	0	0	100	0	0	0
Réunion	99.65900421	100	NAN	NAN	NAN	NAN				
Kuwait	100	100	NAN	NAN	NAN	NAN				
Singapore	100	100	100	0	0	0	100	0	0	0
China, Hong Kong SAR	100	100	100	0	0	0	100	0	0	0
Total										

Our findings

Investigating population size



Most bars are almost entirely basic.

Limited/unimproved/surface are near 0 for most urbanization levels.

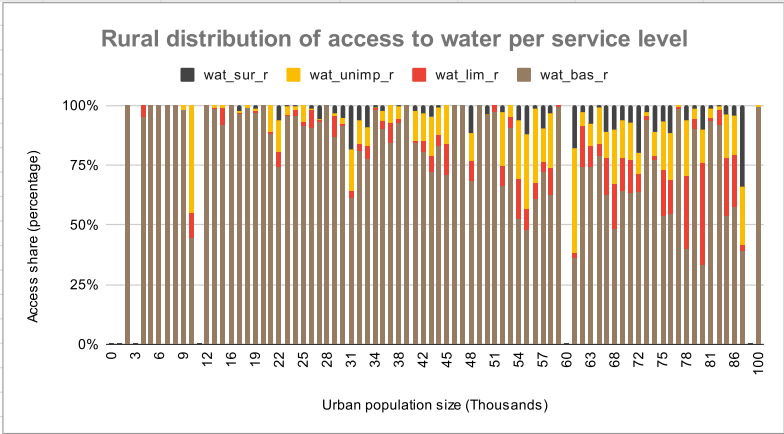
name	pop_r	pop_r (Rounded)	wat_bas_r	wat_lim_r	wat_unimp_r	wat_sur_r	wat_bas_u_clean	wat_lim_u_clean	wat_unimp_u_clean	wat_sur_u_clean
Saint Barthelemy	0	0	NAN	NAN	NAN	NAN				
Nauru	0	0	NAN	NAN	NAN	NAN				
Gibraltar	0	0	NAN	NAN	NAN	NAN				
Saint Martin (French part)	0	0	NAN	NAN	NAN	NAN				
Monaco	0	0	NAN	NAN	NAN	NAN				
Bermuda	0	0	NAN	NAN	NAN	NAN				
China, Macao SAR	0	0	NAN	NAN	NAN	NAN				
Réunion	0.34099579	0	NAN	NAN	NAN	NAN				
Kuwait	0	0	NAN	NAN	NAN	NAN				
Singapore	0	0	NAN	NAN	NAN	NAN				
China, Hong Kong SAR	0	0	NAN	NAN	NAN	NAN				
Qatar	0.76499939	1	NAN	NAN	NAN	NAN				
Guadeloupe	1.50100708	2	NAN	NAN	NAN	NAN				
Belgium	1.92100525	2	100	0	0	0	100	0	0	0
San Marino	2.50099945	3	NAN	NAN	NAN	NAN				
United States Virgin Islands	4.06099701	4	NAN	NAN	NAN	NAN				
Uruguay	4.48500061	4	95.30083	4.69917	0	0	95.30083	4.69917	0	0
Guam	5.06199646	5	NAN	NAN	NAN	NAN				
Malta	5.2559967	5	100	0	0	0	100	0	0	0
Iceland	6.102005	6	100	0	0	0	100	0	0	0
Puerto Rico	6.41899872	6	NAN	NAN	NAN	NAN				
Israel	7.41300201	7	100	0	0	0	100	0	0	0
Northern Mariana Islands	8.20200348	8	NAN	NAN	NAN	NAN				
Netherlands	7.76399994	8	100	0	0	0	100	0	0	0
Argentina	7.88899994	8	NAN	NAN	NAN	NAN				
Japan	8.21800232	8	NAN	NAN	NAN	NAN				
Luxembourg	8.5470047	9	98.5952381	0	1.404761905	0	98.5952381	0	1.404761905	0
Jordan	8.58200073	9	97.31147869	0.6239465945	2.024714716	0.03986	97.31147869	0.6239465945	2.024714716	0.03986
Saint Pierre and Miquelon	10.03800201	10	NAN	NAN	NAN	NAN				
Bahrain	10.49399567	10	NAN	NAN	NAN	NAN				
Gabon	9.90799713	10	44.74588901	10.51114859	44.7429624	NAN				
Martinique	10.86000061	11	NAN	NAN	NAN	NAN				
Lebanon	11.07500458	11	NAN	NAN	NAN	NAN				
Andorra	12.08399963	12	100	0	0	0	100	0	0	0
Denmark	11.88399506	12	100	0	0	0	100	0	0	0
Sweden	12.02300262	12	99.71937474	0	0.2806252626	0	99.71937474	0	0.2806252626	0
Chile	12.27300262	12	100	0	0	0	100	0	0	0
Venezuela (Bolivarian Republic of)	11.72100067	12	NAN	NAN	NAN	NAN				
American Samoa	12.84700012	13	NAN	NAN	NAN	NAN				
Greenland	12.71799469	13	100	0	0	0	100	0	0	0
New Zealand	13.3010025	13	100	0	0	0	100	0	0	0
United Arab Emirates	12.95200348	13	NAN	NAN	NAN	NAN				
Brazil	12.92700195	13	95.94525953	0.9789509154	3.075789551	NAN				
French Guiana	14.18000031	14	NAN	NAN	NAN	NAN				
Oman	13.72400665	14	76.25077991	21.63212113	1.897409218	0.2196897375	76.25077991	21.63212113	1.897409218	0.2196897375
Finland	14.48299408	14	100	0	0	0	100	0	0	0
Australia	13.75900269	14	100	0	0	0	100	0	0	0
Saudi Arabia	15.71299744	16	NAN	NAN	NAN	NAN				
United Kingdom	16.09700012	16	100	0	0	0	100	0	0	0
Norway	17.02600861	17	100	0	0	0	100	0	0	0
Dominican Republic	17.45999908	17	90.30446437	1.385315508	1.115118127	7.195101991	90.30446437	1.385315508	1.115118127	7.195101991
United States of America	17.33599854	17	99.67078734	0	0.3292126628	0	99.67078734	0	0.3292126628	0
Canada	18.43799591	18	99.0604959	0	0.9395041006	0	99.0604959	0	0.9395041006	0

Palau	19.01200104	19	99.75698503	0	0.2430149677	0	99.75698503	0	0.2430149677	0
Costa Rica	19.22900391	19	99.64625	0.35375	0	0	99.64625	0.35375	0	0
Libya	19.30899811	19	NAN	NAN	NAN	NAN				
Spain	19.19000244	19	100	0	0	0	100	0	0	0
Colombia	18.57500458	19	86.76707282	0.7312282016	5.13386897	7.367830005	86.76707282	0.7312282016	5.13386897	7.367830005
Republic of Korea	18.58599854	19	NAN	NAN	NAN	NAN				
France	19.02500153	19	100	0	0	0	100	0	0	0
Mexico	19.26900482	19	98.33706893	0	1.662931067	0	98.33706893	0	1.662931067	0
Greece	20.28499603	20	100	0	0	0	100	0	0	0
Falkland Islands (Malvinas)	21.49200439	21	78.17294282	0	21.82705718	0	78.17294282	0	21.82705718	0
Belarus	20.51699829	21	98.56412231	0.9955971951	0.4402804931	0	98.56412231	0.9955971951	0.4402804931	0
Marshall Islands	22.20600891	22	94.42972	5.39028	0.18	0	94.42972	5.39028	0.18	0
Brunei Darussalam	21.74999237	22	NAN	NAN	NAN	NAN				
Djibouti	21.93800354	22	47.28073293	12.06161014	30.73550468	9.922152249	47.28073293	12.06161014	30.73550468	9.922152249
Peru	21.7029953	22	80.79943849	1.596943771	9.612691163	7.990926578	80.79943849	1.596943771	9.612691163	7.990926578
West Bank and Gaza Strip	23.28100586	23	98.71555642	0.3174006273	0.9670429576	NAN				
Cuba	22.80599976	23	94.38665114	2.591752445	1.905972935	1.115623477	94.38665114	2.591752445	1.905972935	1.115623477
Malaysia	22.84000397	23	90.19744812	0.5444513971	9.258100478	NAN				
Germany	22.54699707	23	100	0	0	0	100	0	0	0
Bulgaria	24.31400299	24	97.40128797	0	2.598712031	0	97.40128797	0	2.598712031	0
Iran (Islamic Republic of)	24.12599945	24	93.82873538	4.297948765	1.602839669	0.2704761905	93.82873538	4.297948765	1.602839669	0.2704761905
Turkey	23.89499664	24	96.02516977	2.684574639	0.7882555911	0.502	96.02516977	2.684574639	0.7882555911	0.502
Cook Islands	24.50499725	25	NAN	NAN	NAN	NAN				
Russian Federation	25.24599457	25	91.54410196	1.583163513	6.872734531	NAN				
Sao Tome and Principe	25.64599609	26	74.30061449	19.71184258	1.729405704	4.258137228	74.30061449	19.71184258	1.729405704	4.258137228
Switzerland	26.08499908	26	100	0	0	0	100	0	0	0
Czech Republic	25.93899536	26	99.8175993	0	0.1824006965	0	99.8175993	0	0.1824006965	0
Algeria	26.26699829	26	90.03753791	8.79672214	0.9946030281	0.1711369258	90.03753791	8.79672214	0.9946030281	0.1711369258
El Salvador	26.55599976	27	93.45439508	0.7804762049	0	5.765128712	93.45439508	0.7804762049	0	5.765128712
New Caledonia	28.48200226	28	NAN	NAN	NAN	NAN				
Hungary	28.05799866	28	100	0	0	0	100	0	0	0
Botswana	29.1230011	29	79.04372386	15.15494718	1.596119156	4.20520981	79.04372386	15.15494718	1.596119156	4.20520981
Iraq	29.10700226	29	94.83114678	2.599251339	0	2.569601877	94.83114678	2.599251339	0	2.569601877
Italy	28.96100616	29	NAN	NAN	NAN	NAN				
Bolivia (Plurinational State of)	29.8769989	30	79.93664283	0.2447364082	3.954133693	15.86448707	79.93664283	0.2447364082	3.954133693	15.86448707
Tunisia	30.43199921	30	93.93623193	3.351687357	2.71208071	0	93.93623193	3.351687357	2.71208071	0
Ukraine	30.39199829	30	100	0	0	0	100	0	0	0
Estonia	30.77099609	31	NAN	NAN	NAN	NAN				
Mongolia	31.34300232	31	61.07075114	3.084170572	17.58078116	18.26429713	61.07075114	3.084170572	17.58078116	18.26429713
Latvia	31.68499756	32	98.57922314	0	1.42077686	0	98.57922314	0	1.42077686	0
Lithuania	31.95400238	32	93.78278023	0	6.217219771	0	93.78278023	0	6.217219771	0
Panama	31.58599854	32	86.32308326	1.791982583	7.162364342	4.722569811	86.32308326	1.791982583	7.162364342	4.722569811
Congo	32.17099762	32	45.72441318	10.6323901	24.40516973	19.23802699	45.72441318	10.6323901	24.40516973	19.23802699
Cabo Verde	33.34799957	33	80.11452544	9.911127022	9.623217003	0.3511305397	80.11452544	9.911127022	9.623217003	0.3511305397
Montenegro	32.51199341	33	98.16188754	0	1.821833386	0.01627906977	98.16188754	0	1.821833386	0.01627906977
Cyprus	33.17900085	33	99.84576172	0	0.1542382836	0	99.84576172	0	0.1542382836	0
Angola	33.17500305	33	27.80822661	8.740488389	22.93315258	40.51813242	27.80822661	8.740488389	22.93315258	40.51813242
South Africa	32.64599609	33	83.32948118	6.977359359	7.76999287	5.916160173	83.32948118	6.977359359	7.76999287	5.916160173
Suriname	33.85099792	34	96.57441263	1.60029586	0.1598017947	1.66548972	96.57441263	1.60029586	0.1598017947	1.66548972
Portugal	33.69000244	34	99.73877609	0	0.2612239122	0	99.73877609	0	0.2612239122	0
Tuvalu	35.98600006	36	100	0	0	0	100	0	0	0
Ireland	36.34700012	36	98.13492075	0	1.865079247	0	98.13492075	0	1.865079247	0
Ecuador	35.83399963	36	87.05074495	0.009631811022	7.26819549	5.671427747	87.05074495	0.009631811022	7.26819549	5.671427747
Morocco	36.46800232	36	77.34612896	12.03707816	6.623063373	3.993729502	77.34612896	12.03707816	6.623063373	3.993729502

Gambia	37.41800308	37	69.18852321	16.49783239	14.24376547	0.0698789368	69.18852321	16.49783239	14.24376547	0.0698789368
Armenia	36.68700027	37	100	0	0	0	100	0	0	0
French Polynesia	38.02499771	38	NAN	NAN	NAN	NAN				
Albania	37.88800049	38	94.09135806	2.305264955	3.603376986	0	94.09135806	2.305264955	3.603376986	0
Paraguay	37.81700516	38	99.11122753	0.8887724684	0	0	99.11122753	0.8887724684	0	0
Democratic People's Republic of Korea	37.61899948	38	88.75362488	0.3564402606	10.18993486	0.7	88.75362488	0.3564402606	10.18993486	0.7
China	38.28691101	38	89.66123351	1.832679086	8.506087409	0	89.66123351	1.832679086	8.506087409	0
Poland	39.95699692	40	100	0	0	0	100	0	0	0
Georgia	40.54700047	41	94.2718842	0	5.703923618	0.02419218345	94.2718842	0	5.703923618	0.02419218345
Nicaragua	40.98799896	41	59.10744663	3.520974285	29.78017368	7.5914054	59.10744663	3.520974285	29.78017368	7.5914054
Austria	41.2519989	41	100	0	0	0	100	0	0	0
North Macedonia	41.51800156	42	97.46871457	1.505202232	0.9105631982	0.11552	97.46871457	1.505202232	0.9105631982	0.11552
Croatia	42.44700241	42	NAN	NAN	NAN	NAN				
Honduras	41.64100266	42	89.87579955	0.8067095079	9.317490946	0	89.87579955	0.8067095079	9.317490946	0
Kazakhstan	42.32899857	42	91.93925234	1.862616064	6.014693049	0.1834385438	91.93925234	1.862616064	6.014693049	0.1834385438
Cameroon	42.43999481	42	43.52758112	12.67259943	30.12380645	13.676013	43.52758112	12.67259943	30.12380645	13.676013
Fiji	42.75299454	43	89.08802405	0	5.759897938	5.152078012	89.08802405	0	5.759897938	5.152078012
Haiti	42.91200256	43	42.8457001	13.25598695	43.89831295	0	42.8457001	13.25598695	43.89831295	0
Ghana	42.65100098	43	71.89275005	11.93454827	4.999356483	11.17334519	71.89275005	11.93454827	4.999356483	11.17334519
Indonesia	43.35900116	43	85.66796238	1.17809123	10.61491462	2.539031772	85.66796238	1.17809123	10.61491462	2.539031772
Kiribati	44.40600204	44	60.99418489	2.096609799	36.90920532	0	60.99418489	2.096609799	36.90920532	0
Jamaica	43.68899918	44	85.39991216	8.490737371	2.568971895	3.540378574	85.39991216	8.490737371	2.568971895	3.540378574
Serbia	43.55400085	44	95.84497475	3.545864028	0.5711602866	0.03800093141	95.84497475	3.545864028	0.5711602866	0.03800093141
Azerbaijan	43.60299683	44	90.92579795	2.391535671	6.682666377	0	90.92579795	2.391535671	6.682666377	0
Slovenia	44.88199615	45	NAN	NAN	NAN	NAN				
Mauritania	44.67300415	45	49.9113389	18.49039689	31.5496173	0.04864690722	49.9113389	18.49039689	31.5496173	0.04864690722
Syrian Arab Republic	44.52499771	45	92.082745	7.917255	0	0	92.082745	7.917255	0	0
Slovakia	46.24000168	46	100	0	0	0	100	0	0	0
Romania	45.80599976	46	100	0	0	0	100	0	0	0
Isle of Man	47.10199738	47	NAN	NAN	NAN	NAN				
Trinidad and Tobago	46.7859993	47	NAN	NAN	NAN	NAN				
Turkmenistan	47.48399734	47	100	0	0	0	100	0	0	0
Namibia	47.96699905	48	71.25616743	11.97218925	7.081803759	9.689839563	71.25616743	11.97218925	7.081803759	9.689839563
Liberia	47.91100311	48	64.10021689	6.537775797	3.383709875	25.97829744	64.10021689	6.537775797	3.383709875	25.97829744
Guatemala	48.16400146	48	90.1193298	1.849169999	4.583438691	3.448061508	90.1193298	1.849169999	4.583438691	3.448061508
Côte d'Ivoire	48.29400253	48	55.72244686	13.34316829	23.17527956	7.759105289	55.72244686	13.34316829	23.17527956	7.759105289
Nigeria	48.04199982	48	61.65821003	7.128924583	20.92487535	10.28799004	61.65821003	7.128924583	20.92487535	10.28799004
Thailand	48.56999969	49	100	0	0	0	100	0	0	0
Uzbekistan	49.58400345	50	96.07258236	0	0	3.927417637	96.07258236	0	0	3.927417637
British Virgin Islands	51.48500061	51	NAN	NAN	NAN	NAN				
Bosnia and Herzegovina	50.97999954	51	97.33333333	2.666666667	0	0	97.33333333	2.666666667	0	0
Benin	51.58500029	52	58.05226799	12.70520504	23.97796833	5.264558634	58.05226799	12.70520504	23.97796833	5.264558634
Senegal	51.87799835	52	75.2429282	4.052382433	20.51245174	0.1922376314	75.2429282	4.052382433	20.51245174	0.1922376314
Philippines	52.59200287	53	91.06193322	3.959520428	4.978546347	0	91.06193322	3.959520428	4.978546347	0
Niue	53.79799652	54	NAN	NAN	NAN	NAN				
Mayotte	54.24900055	54	NAN	NAN	NAN	NAN				
Belize	53.97499847	54	97.99520758	1.358317682	0.6464747393	0	97.99520758	1.358317682	0.6464747393	0
Somalia	53.85900116	54	37.07586282	36.64324958	21.82149546	4.459392143	37.07586282	36.64324958	21.82149546	4.459392143
Democratic Republic of the Congo	54.36199951	54	21.98279234	12.68294146	51.21598167	14.11828453	21.98279234	12.68294146	51.21598167	14.11828453
Zambia	55.37100222	55	48.22728833	8.331703613	31.60865444	11.83235362	48.22728833	8.331703613	31.60865444	11.83235362
Guinea-Bissau	55.80400085	56	49.8319053	9.285773026	40.31645129	0.5658703845	49.8319053	9.285773026	40.31645129	0.5658703845
Mali	56.09099579	56	72.08089521	3.78241223	21.72810084	2.408591717	72.08089521	3.78241223	21.72810084	2.408591717
Republic of Moldova	57.15099716	57	85.48103812	1.52256302	12.99639886	0	85.48103812	1.52256302	12.99639886	0
Sierra Leone	57.07699966	57	52.75270267	5.221136889	23.14294909	18.88321135	52.75270267	5.221136889	23.14294909	18.88321135

Our findings

Investigating population size



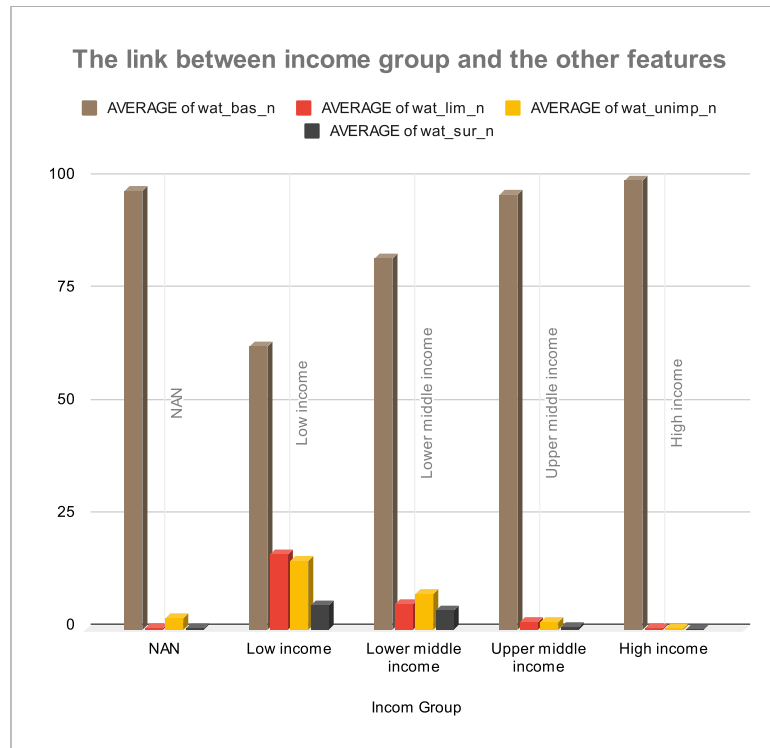
- Much larger shares of limited/unimproved/surface appear compared to urban.

- Rural access varies widely: some bins show near 100% basic, but many bins show sizeable unimproved/surface in some countries.

- The "problem" levels are concentrated in rural areas, and variation is larger across countries.

income_group_n	income_group	SUM of pop_n	AVERAGE of pop_u	AVERAGE of wat_bas_n	AVERAGE of wat_lim_n	AVERAGE of wat_unimp_n	AVERAGE of wat_sur_n
0	NAN	37264.34029	61.46049964	97.1814662	0.1482198143	2.386919485	0.3238794261
1	Low income	590425.1621	36.04465338	62.81937356	16.54838821	15.21093083	5.421307393
2	Lower middle inc	3399310.231	48.78990351	82.21049288	5.688425074	7.898143786	4.285348813
3	Upper middle inc	2547619.307	64.68569565	96.430472	1.561189554	1.483406065	0.5726535552
4	High income	1212076.068	79.35545728	99.56019792	0.1769010855	0.2408458913	0.02205537705

richer income groups tend to be more urbanized. Urbanization often makes it easier to provide piped and improved services at scale, so this can partly explain why water access improves with income (but income remains the strongest driver).



As income group increases, basic access rises, and limited/unimproved/surface decrease sharply.

This means income level is strongly linked to access quality: the “problem” categories are concentrated mainly in low income and partly in lower middle income groups.

The pivot table shows a strong relationship between income group and drinking-water access. Low income countries have the lowest average basic access (≈63%) and the highest shares of limited, unimproved, and surface water. As income rises, basic access increases sharply (≈82% in lower-middle, ≈96% in upper-middle, ≈99.6% in high income), while unsafe categories fall toward zero. This supports the idea that higher national income is associated with stronger water infrastructure and safer service levels.

Executive Summary - Access to Drinking Water (SDG 6) | 2020

Scope	214 countries/areas.
Service levels	At least basic Limited Unimproved Surface water
Area types	National Urban Rural

Key findings

Area gap	Urban access is consistently stronger than rural access. The rural gap is the main driver of deprivation.
Urban (mean across valid countries)	~94.7% at least basic; ~0.3% surface water.
Rural (mean across valid countries)	~81.2% at least basic; ~4.2% surface water (much higher than urban).
Income gradient (national)	Low-income basic ~62.8% vs high-income ~99.6%; unsafe categories approach zero in high-income countries.
Quality assurance	For rows with complete data, the four service levels sum to ~100% (minor floating rounding only).

What was delivered

- Cleaned and validated the JMP export (delimiter inconsistencies; structural checks).
- Engineered population features (urban/rural shares, population binning).
- Produced descriptive statistics and composition visualizations (100% stacked).
- Built an income-group pivot table to quantify the gradient across service levels.